Forcing Diffuse Distributions out of Language Models

Yiming Zhang[1]*, **Avi Schwarzschild**[1], **Nicholas Carlini**[2], **Zico Kolter**[1,3] **& Daphne Ippolito**[1,2]
[1]Carnegie Mellon University
[2]Google DeepMind
[3]Bosch Center for AI

arXiv:2404.10859v2 [cs.CL] 7 Aug 2024

Abstract

Despite being trained specifically to follow user instructions, today's instruction-tuned language models perform poorly when instructed to produce random outputs. For example, when prompted to pick a number uniformly between one and ten Llama-2-13B-chat disproportionately favors the number five, and when tasked with picking a first name at random, Mistral-7B-Instruct chooses Avery 40 times more often than we would expect based on the U.S. population. When these language models are used for real-world tasks where diversity of outputs is crucial, such as language model assisted dataset construction, their inability to produce diffuse distributions over valid choices is a major hurdle. In this work, we propose a fine-tuning method that encourages language models to output distributions that are diffuse over valid outcomes. The methods we introduce generalize across a variety of tasks and distributions and make large language models practical for synthetic dataset generation with little human intervention.[1]

1 Introduction

Consider a Dungeon Master (DM) trying to use a language model based chatbot to assist in managing their Dungeons & Dragons campaign. The DM asks the chatbot to suggest a random name for a character in the story. The first time she asks, it suggests "Anya," and the second time it also suggests "Anya." In fact, almost 40% of the time, the suggested name will be "Anya" even when the language model is deployed with full random sampling. The DM then tries to use the chatbot to roll a twenty-sided die; over 60% of the dice rolls come up as a 14. Frustrated, the DM gives up and brings out their physical dice.

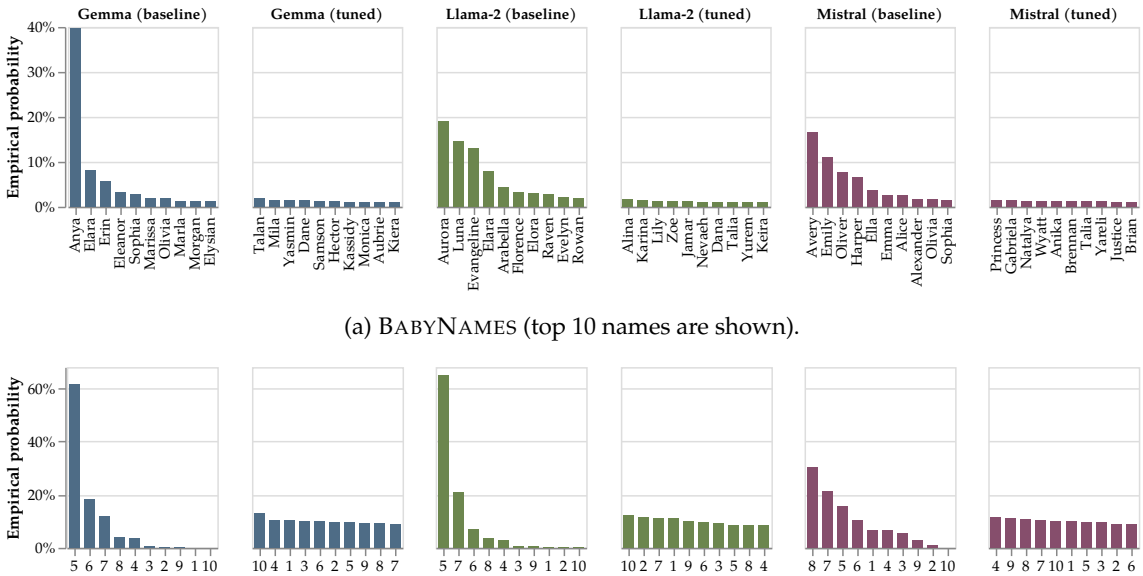
Instruction-tuned language models are extremely bad at producing random outputs when users want them to. Even when prompts are carefully constructed with instructions that encourage randomness, both state-of-the-art open-source and industry language models output very low-entropy distributions over the valid options. Beyond Dungeons & Dragons, there are many practical applications where diversity across valid options is crucial for language model outputs. For example, when language models are used to answer multiple choice or Likert-scale questions, *a priori* each option should be equally likely. When language models are used for synthetic dataset construction, such as for synthetic biographies (Maini et al., 2024; Yuan et al., 2021) or instruction-tuning training sets (Wang et al., 2023), diversity in the generations is essential but arduous to achieve through mere prompt hacking.

In this work, we examine how far off language model generations are from user expectations of randomness and diversity. While our experiments focus on instruction-tuned models such as Llama-2-chat, our findings do generalize to pre-trained language models in general.[2] We then show how

*Correspondence: Yiming Zhang, yimingz3@cs.cmu.edu.

[1]Code and data are available at https://github.com/y0mingzhang/diffuse-distributions.

[2]For example, when GPT-2 (Radford et al., 2019) is prompted with "She rolled the 20-sided die. It landed on the number", the output distribution is similarly far from uniform.



(a) BABYNAMES (top 10 names are shown).

(b) RANDOM NUMBER GENERATION.

Figure 1: **Language models do not produce diffuse probabilities.** The output distributions of untuned Gemma, Llama-2, and Mistral deviate from what we would expect from natural/random distributions. Our tuning method addresses this by diffusing the output distribution over valid candidates. The horizontal axes above are sorted in descending order by probability of the output.

language models can be fine-tuned to produce diffuse distributions over valid options, without sacrificing generation quality. Our method supports tasks where the sample set of valid options is not easily enumerated, and we show that models fine-tuned to produce diffuse probabilities for one set of tasks generalize to other, very different tasks without degrading model utility.

This generalization allows us to promote diversity in complex settings, such as synthetic dataset generation. On the task of generating a dataset of synthetic biographies (similar to the dataset created by Maini et al. (2024) for benchmarking language model unlearning), our method generates four times as many unique first names, three times as many unique birth places, and 1.5 times as many unique careers as the untuned baseline model, all without any need for complex prompt engineering, decoding strategy tweaking or manual re-writing.

2 Preliminaries

We begin with a formal definition of *diffuse probabilities from language models*. We then introduce techniques for measuring diversity of models outputs and we show how to quantify the differences in the observed output distributions and the desired distributions.

2.1 Problem setting

Consider a vocabulary $\mathcal{V} = \{1, 2, ..., n\}$. An autoregressive language model takes a sequence of ℓ tokens $\mathbf{x} \in \mathcal{V}^\ell$ as input and outputs a probability distribution $\mathbf{p}_\theta(\cdot \mid \mathbf{x}) \in \Delta(\mathcal{V})$ over all tokens in the vocabulary. We use $\Delta(\mathcal{V})$ to denote the probability simplex over $\mathcal{V}$.

In generation, we are interested in computing the probability of a potentially multi-token target $\mathbf{y} = [y_1, y_2, \ldots, y_n] \in \mathcal{V}^n$. For example, $\mathbf{y}$ could be a two-digit number or the biography of a person. A language model factors the probability of a sequence $\mathbf{y}$ into the product of probabilities

of individual tokens ($\oplus$ here indicates concatenation):

$$\begin{aligned}\mathbf{p}_\theta(\mathbf{y} \mid \mathbf{x}) &= \mathbf{p}_\theta(y_1 \mid \mathbf{x})\mathbf{p}_\theta(y_2 \mid \mathbf{x} \oplus y_1) \cdots \mathbf{p}_\theta(y_n \mid \mathbf{x} \oplus \mathbf{y}_{<n}) \\ &= \prod_{i=1}^{n} \mathbf{p}_\theta(y_i \mid \mathbf{x} \oplus \mathbf{y}_{<i}). \end{aligned} \tag{1}$$

2.2 Measuring output distributions

How do we measure if an LM produces diffuse probabilities? We consider one method for cases where the sample space is enumerable, and another for when the full sample space is unknown.

In settings such as random number generation within a range, we can enumerate the set of valid generations $\mathcal{T}$. For example, in the context of rolling a twenty-sided die, $\mathcal{T} = \{1, 2, ..., 20\}$, and we expect a perfectly diffuse language model to produce a uniform distribution over $\mathcal{T}$. More generally, for a context $\mathbf{x}$ and a (not necessarily uniform) ground truth probability distribution $\mathbf{p}^\star \in \Delta(\mathcal{V}^\star)$ over generation targets, we can measure the KL-divergence between the model distribution $\mathbf{p}_\theta$ and the ground truth distribution $\mathbf{p}^\star$ as follows:

$$D_{\mathrm{KL}}\left(\mathbf{p}^\star \parallel \mathbf{p}_\theta\right) = \sum_{\mathbf{y} \in \mathcal{T}} \mathbf{p}^\star\left(\mathbf{y} \mid \mathbf{x}\right) \log\left(\frac{\mathbf{p}^\star(\mathbf{y} \mid \mathbf{x})}{\mathbf{p}_\theta(\mathbf{y} \mid \mathbf{x})}\right). \tag{2}$$

KL-divergence measures the difference between the model distribution $\mathbf{p}_\theta$ and the ground truth $\mathbf{p}^\star$, and it attains its minimum at zero if and only if $\mathbf{p}_\theta = \mathbf{p}^\star$.

In many scenarios such as synthetic dataset generation, the support of the ground truth distribution cannot feasibly be enumerated, and we instead measure the *entropy* and the *coverage* of empirical distributions. In these cases, we sample a large number of generations $\mathbf{y}_1, \ldots, \mathbf{y}_N$ from the model and use the empirical distribution $\tilde{\mathbf{p}}$ derived from generations to approximate $\mathbf{p}_\theta$. Entropy of the empirical distribution quantifies the diversity of model generation:

$$\mathrm{Entropy}(\tilde{\mathbf{p}}) = -\sum_{\mathbf{y} \in \mathcal{V}^\star} \tilde{\mathbf{p}}(\mathbf{y}) \log \tilde{\mathbf{p}}(\mathbf{y}). \tag{3}$$

Coverage-N is defined as the number of unique generations produced by the model in N samples:

$$\text{Coverage-}N(\tilde{\mathbf{p}}) = |\{\mathbf{y}_i \mid i \in [N]\}|. \tag{4}$$

For a sufficiently large N, coverage quantifies the number of unique generations a user interacting with the system could practically observe.

3 Forcing Diffuse Probabilities

We introduce a method for fine-tuning language models to encourage diffuse probabilities distributions. Our technique hinges on the fact that fine-tuned models generalize—a model which is trained to produce diffuse probabilities for one task will also do so for tasks unseen during training.

3.1 Distribution matching as language model fine-tuning

Suppose, for some task, we already know the target set $\mathcal{T}$ of possible outputs, and the ground truth distribution $\mathbf{p}^\star$ we would like the language model to produce. Our method simply maximizes the sum of model log-likelihood over all targets, weighted by their ground truth likelihoods:

$$\mathcal{L}(\mathbf{p}_\theta) = -\sum_{\mathbf{y} \in \mathcal{T}} \mathbf{p}^\star(\mathbf{y} \mid \mathbf{x}) \log \mathbf{p}_\theta(\mathbf{y} \mid \mathbf{x}). \tag{5}$$

This objective is, in fact, the cross-entropy loss ubiquitous in language modeling (Berger et al., 1996) extended to multiple target sequences (Edunov et al., 2018). Since the model distribution $\mathbf{p}_\theta$